

Server.app Reverse Proxy 1.0.10

In Mac OS X Server 10.6 Snow Leopard it was relatively easy to run third party server applications like FileMaker Server, Kerio Connect, and Rumpus alongside the OS X Server Web/Wiki.

But starting in 10.7 and then continuing in 10.8 through 10.11, Server.app runs an instance of Apache (httpd) even when you are NOT using Web/Wiki or even Profile Manager and this made it impossible to install FileMaker Server 13 on the same box as OS X Server. While FileMaker Server 14 brings that ability back it is by moving FileMaker Server to different ports entirely.

This article will show how to use Server.app's reverse proxy features to install and run FileMaker Server, Kerio Connect, and Rumpus all the while allowing them to use port 80 and 443 alongside the Apple services you love like Web, Wiki and Profile Manager in a way that will survive OS X .x updates because it takes advantage of the Web App feature in Server.app (which is now the recommended way of modifying OS X Apache.*

***Not saying it is "Best Practices", nor that we recommend it; just saying you can do it.**

What am I talking about? Well, let's take a look.

Download Server.app 3 to your Mavericks, or Server.app 5.0.15 or higher to your Yosemite or El Capitan Mac and launch it. Set up Open Directory service so you can use a VPN (supposedly a non-web service). Do a port scan for port 80 or 443 and you will see that nothing is running yet.

In fact, Server.app 5's OD is going to start Apache listening on 80, 443, 8008, 8800, 8443, and 8843!

In Server 3 you only needed to turn on the VPN service. Whammo! The Apache daemon located in /usr/sbin/httpd starts running taking up port 80 (fig. 1) and 443 and making it impossible for FileMaker Server 13 to work or even install properly. But with Server 5 as soon as you have OD going ALL the httpd ports are being listened on by Server app.

In the interest of making this tutorial still relevant to Server app 3 and 4 I will keep the original working mostly intact and ask that if you have Server app you review Appendix C carefully.

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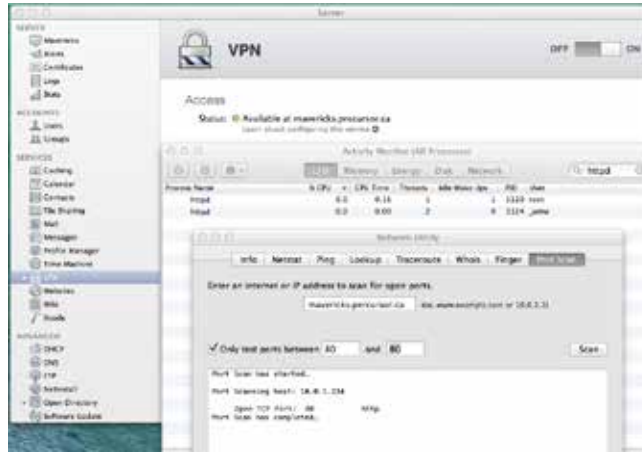


Fig 1. In Server app 3m Apache is running even though Web/Wiki/Profile Manager are off

So, we are going to:

- install Server.app and provide DNS
- set up Open Directory
- set up our Reverse Proxies
- temporarily disabled all instances of Apache
- install FMS, Kerio Connect, and Rumpus and configure to special ports
- turn on Apache again.

First lets get our services, dns and ports sorted out (fig. 2):

Server	FQDN	HTTP	HTTPS
OS X Server.app	osxserver.precursor.ca	80	443
Maxum Rumpus 8	rumpus.precursor.ca	8000	8010
FileMaker Server 14	filemaker.precursor.ca	8001	8011
Kerio Connect 8.5	kerio.precursor.ca	8003	8013

Fig 2. DNS and port assumptions

1. Purchase and install Server.app from the App Store. (For Yosemite and El Capitan use v.5.0.15 or higher).
2. Set up DNS. I am going to assume you know what you are doing here as it is beyond the scope of this tutorial to deal with this. You should have an A record for the Fully Qualified Domain Name (FQDN) of your server: *osxserver.precursor.ca*

And you should have aliases (CNAMEs) for rumpus, filemaker, and kerio (as per figure 2 above).

3. Set up an Open Directory but turning it on for the FQDN of your server: *osxserver.precursor.ca*
4. Turn on VPN service (this will turn on instance of Apache that cannot be controlled by the GUI Web or Wiki controls). DO NOT turn on Profile Manager yet –or even look at its settings since if you breath on it you will be turning on Apache there too!

Do a port scan with Network Utility (/System/Library/CoreServices/Applications/Network Utility.app) and you will see port 80 is in use.

5. Now temporarily turn OFF this hidden Apache using Terminal.

```
sudo launchctl unload -w /System/Library/LaunchDaemons/org.apache.httpd.plist
```

You can do a port scan with Network utility to verify port 80 is no longer in use.

6. Now we are going to install our 3rd party servers.

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Rumpus

Lets start with Rumpus as an example since it defaults to port 8000 for web.

Download Rumpus from www.maxum.com and place it in your Applications folder.

Launch Rumpus and in its SetUp Assistant step 3 Enable the WFM for port 8000:



Fig 3. Enable WFM on port 8000 in Rumpus

You may also want to enable secure web services for Rumpus. This is done in the Network Settings: Secure Services tab. For this example install we chose port 8010 as the HTTPS port.

You will also want to tell Rumpus that it is going to be running behind a reverse proxy. Do so by choosing "Open Config Folder" from the "File" menu. Then modify the "Rumpus.conf" file to include the line:

```
WFMThruProxy YES
```

Then save the file, and stop and restart Rumpus.

We assume you can deal with certificates here either self-signed or commercial – In Appendix A you will some info about moving certificates between OS X and these apps.

FileMaker Server

That was easy. Now let's do FileMaker Server.

In a previous version of this tutorial the next step was to configure Kerio Connect. However, the current versions of Kerio Connect and FileMaker Server use two different versions of Java to complete the installation. Therefore, we will install FileMaker Server first since FileMaker Server 13.05 utilizes Java 7 and Kerio Connect 8.4 will force you to install Java JDK 8.

First, use Apple's Network Utility to do a port scan to make certain that port 80 and port 443 are not in use.

Once you are sure that port 80 and 443 are not being used go ahead and install FileMaker Server 13.05.

It is beyond the scope of this document to go into great detail on the stock installation of FileMaker Server. I assume you can install it using the dns naming convention from figure 2 above.

NB. The FileMaker Server Admin Console name ("server Name" in the General Settings tab of Admin Console) must be identical to the Apple System Preferences: Sharing name. (FileMaker Answer ID 14127).

FileMaker Server 14

FileMaker Server 14 is easy to work with as its installer allows you to specify alternate ports during the installation procedure. Then all we have to do is make a couple of small adjustments to the config files.

Since FileMaker Server 14 is the currently shipping product lets start with it.

When the install is completed you will want to make sure that deal with FileMaker Answer ID 14127 (above).

Then there are just two things to modify in the config files:

httpd.conf

/Library/FileMaker Server/HTTPServer/conf/httpd.conf

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Look for the line starting with:

```
#Various Default Settings
```

And uncomment the following line (remove the "#") so that it reads:

```
Include conf/extra/httpd-default.conf
```

httpd-default.conf

/Library/FileMaker Server/HTTPServer/conf/extra/httpd-default.conf

If you are using Yosemite (which comes with Apache 2.4) then in this file you must change:

```
UseCanonicalName Off
```

to:

```
UseCanonicalName On
```

FileMaker Server 13

A lot more massaging of config files is required in these older versions.

Once you have installed FileMaker Server 13 you can stop it in the console because we have to adjust the ports manually.

So go ahead and log in to the FileMaker Server console at <http://localhost:16001/admin-console>.

You can stop the Database server and the Web Publishing engine.

There are several text files we are going to have to edit.

I like using BBEdit or its free counterpart TextWrangler to do so.

httpdctl

/Library/FileMaker Server/HTTPServer/bin/httpdctl

in this file edit the line beginning with "SERVER_NAME" to include your FQDN (from figure 2 above).

Of course you will substitute your own domain name for "precursor.ca"

```
SERVER_NAME=filemaker.precursor.ca
```

httpd.conf

/Library/FileMaker Server/HTTPServer/conf/httpd.conf

in this file edit the listening port from 80 to 8001 (as per figure 2 above).

```
Listen 8001
```

And use that port to edit the server name:

```
ServerName filemaker.precursor.ca:8001
```

If you are using Yosemite server (which comes with Apache 2.4) with FMS 13 then you must also edit this:

Look for the line starting with:

```
#Various Default Settings
```

And uncomment the following line (remove the "#") so that it reads:

```
Include conf/extra/httpd-default.conf
```

httpd-ssl.conf

/Library/FileMaker Server/HTTPServer/conf/extra/httpd-ssl.conf

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in this file you will make three edits:

```
Listen 8011
<VirtualHost _default_:8011>
SERVER_NAME=filemaker.precursor.ca
```

httpd-rewrite-fmiwebd.conf

/Library/FileMaker Server/HTTPServer/conf/extra/httpd-rewrite-fmiwebd.conf

in this file you will make this edit:

```
RewriteRule ^/fmi/webd https://%{HTTP_HOST}:8011%{REQUEST_URI}
```

httpd-default.conf

/Library/FileMaker Server/HTTPServer/conf/extra/httpd-default.conf

If you are using Yosemite (which comes with Apache 2.4) with FMS 13 then in this file you must change:

```
UseCanonicalName Off
```

to:

```
UseCanonicalName On
```

Once you have made and saved all these changes you may log in to the FileMaker Server Console and start the Database and the Web Publishing Engine again.

Kerio Connect

Download Kerio Connect and the Kerio Connect Open Directory Plugin.

Install the Open Directory Plugin first. Then install Kerio Connect (note you need the Java JDK and not the Java JRE which got installed with FileMaker. You may also use Apple's OS X Java for Mac 2015).

Since OS X Server 5 even only running OD will be listening on ports 80, 443, 8008, 8800, 8443 and 8843. We will reset Kerio's HTTP and HTTPS services to use ports 8003 and 8013 respectively.

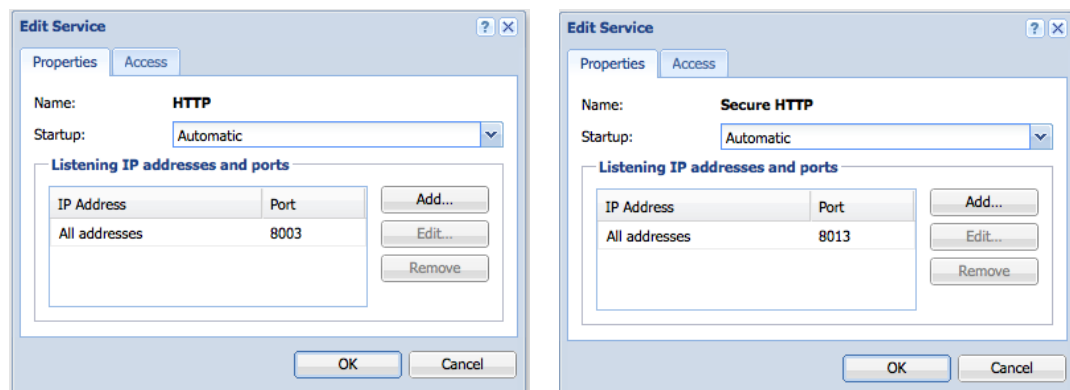


Fig 4 a & b. Turning off default ports of 80, 8800 and 443, 8843 in Kerio Connect

Incidentally, you will also want to edit the ports for LDAP and Secure LDAP from their default values. This will avoid error messages as the Kerio services cannot run on the same ports as the OS X Server LDAP which was activated earlier when we turned on Open Directory. For this we can just add 99 to the end of the number.

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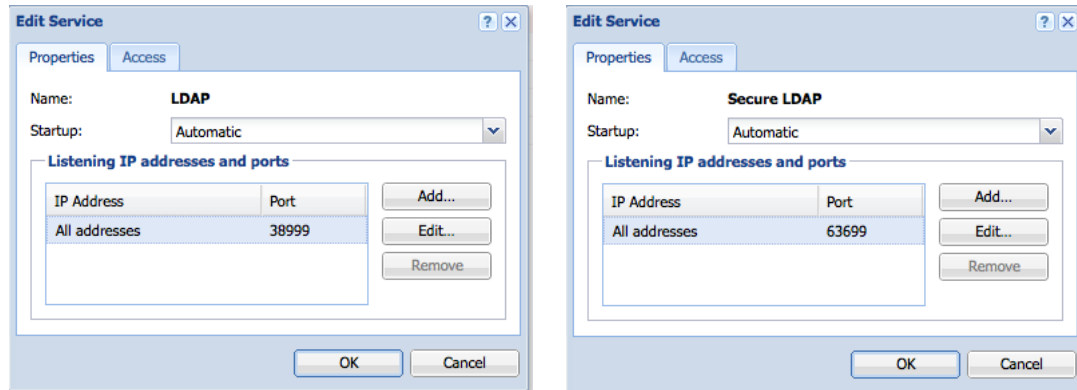


Fig 5 a & b. Editing the LDAP and Secure LDAP ports in Kerio Connect

Use Network Utility and test to make sure that your port assignments are working and the ports 8001 and 8011 are active and that ports 80 and 443 are not.

If you have succeeded you are now ready to set up OS X Apache with the reverse proxies.

To begin with you can now enable to hidden instance of Apache before we forget.

In Terminal:

```
sudo launchctl load -w /System/Library/LaunchDaemons/org.apache.httpd.plist
```

This will allow OS X Server.app services like Web, Wiki, Profile Manager and VPN to work properly.*

*** NB** that this will work in Server app v3 and v4 but the Read Me for Server v.5 specifically states:

" Other instances of Apache are typically running to support the proxy but are not accessibly by any apachectl command." So it may not be possible to install Server first and turn off apache instances. You may have to install your Kerio, FileMaker, and Rumpus first and change their ports before installing Server.app

The OS X Server.app Web process has a GUI but you need to do a little customization work to get your Reverse Proxies to show in that GUI.

So before we start the web process lets do the nitty gritty editing. We are going to make Apache web apps. It is not something well documented on Apple's web site or even developer.apple.com although you can read the man pages in Terminal for *webappctl* and *webapp.plist*.

We are going to make some text files for:

/Library/Server/Web/Config/apache2/

for example we will need two files for the HTTP of a web-site and two more for the HTTPS or a web site.

Let's start with Rumpus which we assigned HTTP to port 8000 and HTTPS to port 8010 (see figure 2).

For the HTTP port 8000 we will need:

/Library/Server/Web/Config/apache2/httpd_rumpuswebapp.conf

/Library/Server/Web/Config/apache2/ca.precursor.rumpuswebapp.plist

httpd_rumpuswebapp.conf

(I used bold text for emphasis)

```
# from http://httpd.apache.org/docs/2.2/mod/mod_proxy.html#proxypass
```

```
RewriteEngine On
```

```
RewriteCond %{HTTPS} =off
```

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```

RewriteRule . - [E=protocol:http,E=port:8000]
RewriteCond %{HTTPS} =on
RewriteRule . - [E=protocol:https,E=port:8010]
ProxyPassReverse / http://rumpus.precursor.ca:8000/
ProxyPass / http://rumpus.precursor.ca:8000/

```

Save a text file with the above contents into */Library/Server/Web/Config/apache2/httpd_rumpuswebapp.conf*

Now you are ready for stage 2, the plist.

ca.precursor.rumpuswebapp.plist

Save a text file with the contents below into:

/Library/Server/Web/Config/apache2/webapps/ca.precursor.rumpuswebapp.plist

Contents of plist (Again, I have bolded for emphasis):

```

<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">

<!-- See man pages for webapp.plist(5) and webappctl(8) for information about this example webapp.plist -->

<plist version="1.0">
<dict>
    <key>includeFiles</key>
        <array>          <!-- Include files are activated in virtual host when webapp is started -->
            <string>/Library/Server/Web/Config/apache2/httpd_rumpuswebapp.conf</string>
        </array>
    <key>name</key>
    <string>ca.precursor.rumpuswebapp</string>
    <key>displayName</key>          <!-- Name shown in Server app -->
    <string>RumpusWebApp</string>
    <key>installationIndicatorFilePath</key><!-- The presence of this file indicates web app is installed -->
    <string>/usr/local/Rumpus/rumpus.conf</string>
    <key>sslPolicy</key><!-- Determines webapp SSL behavior -->
    <integer>0</integer><!-- 0: default, UseSSLWhenEnabled -->
                                <!-- 1:          UseSSLAlways -->
                                <!-- 2:          UseSSLOnlyWhenCertificateIsTrustable -->
                                <!-- 3:          UseSSLNever -->
                                <!-- 4:          UseSSLAndNonSSL -->

```

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```
</dict>
</plist>
```

The permissions for these files need to be owner: root and group: wheel, 755.

You can easily apply this in Terminal:

```
sudo chown -R root:wheel /Library/Server/Web/Config/apache2/httpd_rumpuswebapp.conf
sudo chmod -R 755 /Library/Server/Web/Config/apache2/httpd_rumpuswebapp.conf
sudo chown -R root:wheel /Library/Server/Web/Config/apache2/webapps/httpd_rumpuswebapp.plist
sudo chmod -R 755 /Library/Server/Web/Config/apache2/webapps/httpd_rumpuswebapp.plist
```

but I like to use the GUI of the utility Batchmod which is available at:

<http://www.lagentesoft.com/Batchmod/>

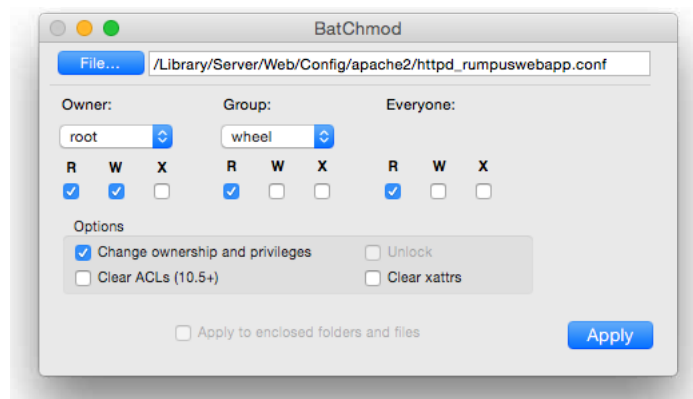


Fig 6a. Using BatchMod to modify permissions

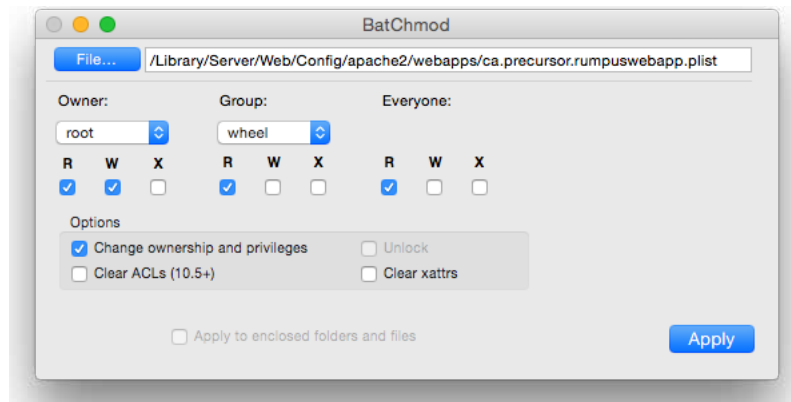


Fig 6b. Using BatchMod to modify permissions

We will make this type of web app file set for each instance of http and https that we need to reverse proxy.

And we will have to change the ownership and permissions for each to root:wheel 755.

So now let's take a look at the Rumpus SSL files:

httpd_rumpusSSLwebapp.conf

from http://httpd.apache.org/docs/2.2/mod/mod_proxy.html#proxypass

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RewriteEngine On

```
RewriteCond %{HTTPS} =off
RewriteRule . - [E=protocol:http,E=port:8000]
RewriteCond %{HTTPS} =on
RewriteRule . - [E=protocol:https,E=port:8010]
ProxyPassReverse / https://rumpus.precursor.ca:8010/
ProxyPass / https://rumpus.precursor.ca:8010/
```

Save a text file with the above contents into `/Library/Server/Web/Config/apache2/httpd_rumpusSSLwebapp.conf` and make sure to change the ownership and permissions accordingly using either Terminal or Batchmod.

ca.precursor.rumpusSSLwebapp.plist

Save a text file with the contents below into:

`/Library/Server/Web/Config/apache2/webapps/ca.precursor.rumpusSSLwebapp.plist`

```
<?xml version="1.0" encoding="UTF-8"?>
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">
<!-- See man pages for webapp.plist(5) and webappctl(8) for information about this example webapp.plist -->
<plist version="1.0">
<dict>
    <key>includeFiles</key>
    <array>
        <!-- Include files are activated in virtual host when webapp is started -->
        <string>/Library/Server/Web/Config/apache2/httpd_rumpusSSLwebapp.conf</string>
    </array>
    <key>name</key>
    <string>ca.precursor.rumpusSSLwebapp</string>
    <key>displayName</key>
    <!-- Name shown in Server app -->
    <string>RumpusSSLWebApp</string>
    <key>installationIndicatorFilePath</key>
    <!-- The presence of this file indicates web app is installed -->
    <string>/usr/local/Rumpus/rumpus.conf</string>
    <key>sslPolicy</key>
    <!-- Determines webapp SSL behavior -->
    <integer>0</integer>
    <!-- 0: default, UseSSLWhenEnabled -->
    <!-- 1: UseSSLAlways -->
    <!-- 2: UseSSLOnlyWhenCertificateIsTrustable -->
    <!-- 3: UseSSLNever -->
    <!-- 4: UseSSLAndNonSSL -->
</dict>
</plist>
```

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The Kerio and FileMaker files are going to be similar except you will use the ports and DNS names outlined in figure 2 above. (**NB.** see the included example files for FileMaker Server console proxies)

For the FileMaker Server <String> use:

/Library/FileMaker Server/HTTPServer/conf/httpd.conf

And for the config <string> for Kerio you will use its default location:

/usr/local/kerio/mailserver/mailserver.cfg

If your Kerio mail store has been moved to an external array then of course you would modify as necessary.

To make it easier (i.e. to cut and paste) a complete set of these files is included as a resource in the download of this tutorial at the rais.precursor.ca site. Just substitute your domain wherever you see "precursor.ca" or "ca.precursor" as the case may be. Once you have created and saved all your config and plist files and set their ownership and permissions correctly, restart your computer so that they will get picked up in the GUI in the next step.

Enable Web Sites and use the GUI to start the reverse proxies!

Lets start with Rumpus again. First, enable the Web service in Server.app.

Then create your Rumpus web site: rumpus.precursor.ca and leave it set to port 80.

Click the *Edit Advanced Settings...* button:

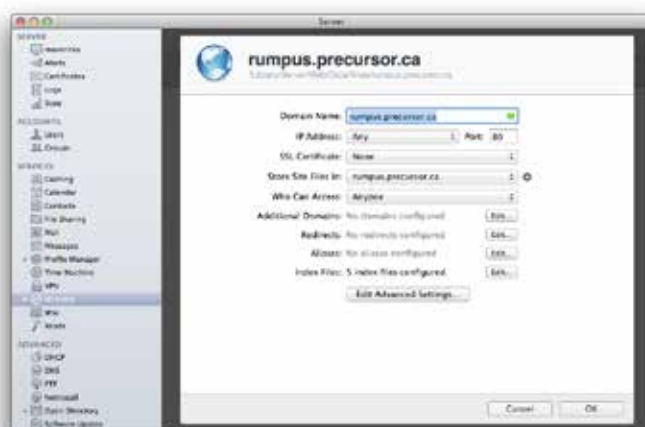


Fig 7. The Advanced Settings button

Now enable the WebApp by putting a checkbox beside *RumpusWebApp*

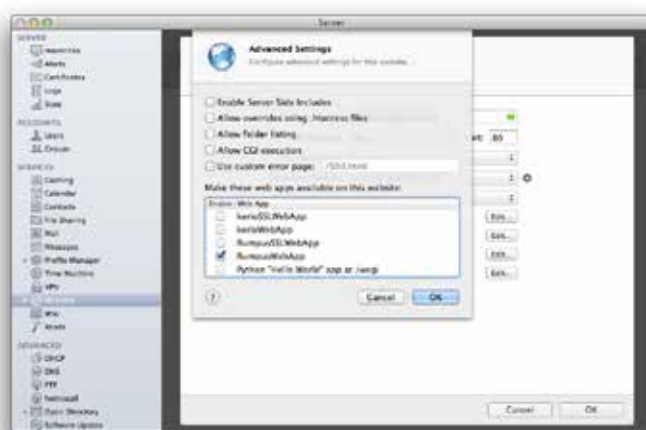


Fig 8. Enable the RumpusWebApp

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You will now do the same thing for the HTTP Port 80 sites for FileMaker Server and Kerio Connect

And you will also create the SSL web site for each of these.

Starting with Rumpus again:

Create your Rumpus SSL web site <rumpus.precursor.ca> but set it to port 443 and in the Advanced Settings use the *rumpusSSLwebapp*.

Do the same thing for the Kerio SSL and FileMaker Server SSL sites.

You should now have OS X Server Web, Wiki, and Profile Manager all properly using ports 80 and 443 and you should have the following reverse proxies in effect:

```
http://rumpus.precursor.ca -> http://rumpus.precursor.ca:8000
https://rumpus.precursor.ca -> https://rumpus.precursor.ca:8010
```

```
http://kerio.precursor.ca -> http://kerio.precursor.ca:8003
https://kerio.precursor.ca -> https://kerio.precursor.ca:8013
```

```
http://filemaker.precursor.ca -> http://kerio.precursor.ca:8001
https://filemaker.precursor.ca -> https://kerio.precursor.ca:8011
```

You will also have the following functional URLs:

OS X Server: <http://osxserver.precursor.ca>

FileMaker console: <http://localhost:16001/admin-console>

Kerio Connect Admin: <http://localhost:4040> or <https://kerio.precursor.ca:4040>

And, if you choose to enable these services in Server.app:

OS X Server Wiki: <http://osxserver.precursor.ca/wiki>

OS X Server Profile Manager: <https://osxserver.precursor.ca/profilemanager>

Additional Resources

Kai Howells of the Australian company Automatica has written an AppleScript to generate the Kerio web app and can be edited to do the others: <https://github.com/kai-h/Kerio-Connect-Webapp-Configurator>

Automatica is at: <http://automatica.com.au/>

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Revision History:

- 1.0.10 October 24/15. Corrected for OS X Server 5.0.15 fixes and revised instructions for v.3.x and v.4.x to be inline with v.5.0.15.
- 1.0.9 October 10/15. Corrected typos in Appendix C: Server 5 and in the kerio example files.
- 1.0.8 October 1/15. Revised with a working solution for the recently Server app 5.
 Changed ports for Kerio to 8003 nad 8013 to avoid conflicts with Server 5,
 Renamed exmple files from "fms13" to "filemaker" for clarity.
 Detailed Info on OS X Server 5 appears in newly added Appendix C.
- 1.0.7 July 14/15. Added an Appendix B to cover Reverse Proxy to a second internal server
- 1.0.6 May 31/15. Now includes Instructions for FileMaker Server 14.
- 1.0.5 May 11/15. Corrections in the certificates appendix and corrections of typos in the example files.
- 1.0.4 January 24/15, Important config info about using commercial certs (GoDaddy etc); Fixed typo in httpd_Keriowebapp.conf example file;
- 1.0.3 December 30/14, Re-edited for Yosemite and Server.app 4. Added file ownership and permission info.
- 1.0.2 September 29/14, Added Appendix A with info on importing/Exporting SSL certificates.
- 1.0.1 July 22/14, Add reference to Automatica's AppleScript
- 1.0 July 19/14, Show how to set up reverse proxies for Rumpus, Kerio Connect, and FileMaker Server 13 on OS Mavericks with Server.app 3.

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Appendix A: Import/Export SSL certificates

It should be obvious that if you are reverse proxying through Apple's web server the service you use behind that must be using the same SSL Security Certificate as the the one you use in OS X Server.app.

In some cases you may have been using the service (e.g. Kerio Connect) before doing reverse proxy – in which case you will have applied for the certificate within Kerio Connect and need to export it to OS X.

However, if you are starting from scratch it may be easier to apply for, and install your SSL certificate in Server.app and then export it to the background service (again I will use Kerio as the example).

Exporting from OS X to Kerio Connect or FileMaker Server

Export a .p12 file from Keychain Access

Open Keychain Access

Select "System" and click the Lock icon to unlock it supplying the admin password for your computer.

In Category, select Certificates and find your mail server cert. Select it.

Right Click and choose Export "*kerio.precursor.ca*"...

Choose the File Format : Personal Information Exchange (.p12).

Now, use Terminal to massage the constituent keys you need out of the .p12 file you just exported.

```
openssl pkcs12 -in path.p12 -out kerio.precursor.ca.crt.pem -clcerts -nokeys
-clcerts only outputs client certificates (not CA certificates)
-nokeys no private keys are output
```

```
openssl pkcs12 -in path.p12 -out kerio.precursor.ca.key.pem -nocerts -nodes
-nocerts no certificates at all will be output
-nodes don't encrypt the private keys at all
```

In the Finder you can remove the .pem from the above names.

Finally, import the keys into Kerio Connect by going into the *SSL Certificates* pane and clicking the Import button and select the .key and the .crt files into their appropriate fields.

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Importing into OS X from Kerio Connect

To export you have to be logged in as the root user so you would have to *Enable Root* in Directory Utility from / System/Library/CoreServices/Applications. Once logged in as root,

In Kerio Connect's *SSL Certificates* pane select the Active Certificate

Export the certificate and the private key files using the *Export* button. You should have:

kerio.precursor.ca.crt and *kerio.precursor.ca.key*.

Use the certificate and key files to make a .p12 file which you can import into OS X Keychain access:

```
openssl pkcs12 -export -clcerts -inkey kerio.precursor.ca.key -in kerio.precursor.ca.crt -out kerio.precursor.ca.p12
```

-clcerts only outputs client certificates (not CA certificates).

Enter Export password: I use login password

Then Verify password:

In Keychain Access select the System Keychain and unlock it using the login password.

Select the Certificates category, then

From the File menu select *Import Items...* and navigate to the .p12 file you made above into the destination *Certificates*. Supply the export password you made earlier.

Third party certificates will not work in OS X Server.app without following the tech note below...

Notes.

1) For all of the above you will supply the full path the file into terminal. You can get this by typing it out manually or be dragging the file or the enclosing folder into the Terminal Window.

2) In all case you will substitute your own server FQDN for "*kerio.precursor.ca*"

Using 3rd party Commercial Certificates (GoDaddy, Comodo, GeoTrust, etc.)

OS X Server.app's Web Services panel won't let you create SSL websites with 3rd party certs properly unless you deal with a Keychain Access control issue.

See this important Apple Knowledgebase article if you are having trouble selecting port 443 when you choose a 3rd party commercial cert in the GUI:

<http://support.apple.com/en-ca/HT203731>

OS X Server: Access Controls might prevent a certificate identity from working with Server services

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Appendix B: 2 Computer Reverse Proxy

The reverse proxy instructions were for one computer housing both a functioning OS X Server and other servers like Kerio Connect, FileMaker Server and Rumpus. But you may want to have your OS X Server on one machine using the port forwards for port 80 and 443 and put other servers like Kerio on other machines but still use the default ports 80 and 443.

Exporting from OS X to Kerio Connect or FileMaker Server

Instead of using an FQDN (Fully Qualified Domain Name) like "kerio.precursor.ca" in the Proxy directive you will use the internal address of the second computer where the Kerio Connect server resides (in the example I use 1.2.3.4).

The port 80 web app < httpd_Keriowebapp.conf> will need to include these lines (where the i.p. address of the Kerio server is substituted for "1.2.3.4"):

```
ProxyPreserveHost On
ProxyPassReverse / http://1.2.3.4:80/
ProxyPass / http://1.2.3.4:80/
ServerName kerio.precursor.ca
```

PreserveHost keeps the name in the URL and ServerName tells which name to keep there.

The port 443 web app < httpd_KerioSSLwebapp.conf> will need to include these lines (where the i.p. address of the Kerio server is substituted for "1.2.3.4"):

```
SSLProxyEngine on
SSLProxyVerify none
SSLProxyCheckPeerCN off
SSLProxyCheckPeerName off
SSLProxyCheckPeerExpire off
ProxyPreserveHost On
ProxyPassReverse / http://1.2.3.4:443/
ProxyPass / http://1.2.3.4:443/
ServerName kerio.precursor.ca
```

PreserveHost keeps the name in the URL and ServerName tells which name to keep there.

Turning off ProxyVerify, ProxyCheckPeerCN, ProxyCheckPeerName, and ProxyCheckPeerExpire will prevent the client browser from getting certificate errors.

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Appendix C: OS X Server 5

Apple's new OS X Server v.5 is now OS independent and works on both Yosemite 10.10 and El Capitan 10.11. Great! But it also brings some fairly big changes in Reverse Proxy activity. Although Apple fixed some problems with version 5.0.15 users should check out the ReadMe in /Library/Server/Web/Config/apache2/ReadMe.txt (reprinted here with spelling mistakes intact):

Special notes about the web proxy architecture in Server application 5.0:

This version of Server application contains a revised architecture for all HTTP-based services. In previous versions there was a single instance of httpd acting as a reverse proxy for Wiki, Profile, and Calendar/Address services, and also acting as the Websites service. With this version, there is a major change: A single instance of httpd runs as a reverse proxy, called the ServerProxy, and several additional instances of httpd run behind that proxy to support specific HTTP-based services, including an instance for the Websites service.

Since the httpd instance for the Websites service is now behind a reverse proxy, or ServerProxy, note the following:

- /usr/sbin/apachectl controls desktop Apache by loading/unloading the org.apache.httpd launchd job; it should not be used when Server app is installed.
- /Applications/Server.app/Contents/ServerRoot/usr/sbin/server-apachectl controls Server application's Website Apache by enabling/disabling the com.apple.server.httpd launchd job using serverctl.
- **Other instances of Apache are typically running to support the proxy but are not accessible by any apachectl command.**
- The Websites service httpd instance listens for TCP traffic on:
 - localhost, on special 34580 and 34543 TCP ports, for default sites and for custom sites configured with standard ports 80 or 443. These are behind the proxy.
 - External IP addresses, on TCP ports other than 80 and 443, for custom sites configured with non-standard ports. These are not behind the proxy.
- It is only the external ServerProxy httpd instance that listens on TCP ports 80 and 443; it proxies HTTP requests and responses to Websites and other HTTP-based services.
- The external ServerProxy httpd instance also listens on TCP ports 8008, 8800, 8443, and 8843 to support the Calendar & Contacts service. Attempting to bind these ports in other services will cause a conflict.
- Adjustments have been made to the directives in Websites service config files:
 - The Listen directives specify 127.0.0.1; only the ServerProxy listens on an external IP address.
 - The VirtualHost directives specify the internal address and port, so for the default sites it's 127.0.0.1:34580 (for the default site) and 127.0.0.1:34543 (for the default SSL site).
 - SSL Certificates are configured in the ServerProxy config file, not in the site config files.

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The ServerProxy is the SSL endpoint, and the TCP traffic between the proxy and the local-host Websites httpd is not encrypted.

- To allow correct construction of redirects through the proxy, a ServerName directive is present in virtual host config files. "ServerName default" is present for wild-card default sites, and "ServerName http[s]://vhostname:nnn" is present in custom sites. For the latter, the scheme and port number nnn are the external ones, not the internal ones.
- CustomLog directives that correctly log external IP addresses are in use.
- Directives that rely on remote IP addresses for correct operation, such as Allow/Deny, Require ip, and also CGIs that use the REMOTE_ADDR variable should operate correctly since mod_remoteip is configured by default.
- The ServerProxy keeps its own set of logs:
 - /var/log/apache2/service_proxy_access.log
 - /var/log/apache2/service_proxy_error.log
- Additional service-specific instances of httpd keep their own logs:
 - /var/log/apache2/services/*_log

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